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Predicting the Lightness of Chromatic Object Colors Using CIELAB

A metric of perceived lightness of object colors, L**, *is derived and evaluated. This metric, based on CIELAB parameters, is designed to predict the Helmholtz–Kohlrausch effect whereby chromatic object colors appear lighter than achromatic object colors of the same luminance factor. Models were fitted to published data and then tested with results from a new lightness-matching experiment that is described in this article. The* L** *metric predicts perceived lightness to within the interobserver variability and provides a simple, practical measure of this appearance attribute.*

INTRODUCTION

Lightness is defined as the brightness of an area judged relative to a similarly illuminated area that appears white or highly transmitting.[1] The definition requires that a color must be viewed as a related color in order to make judgements of lightness. Object colors are usually also related colors, therefore it is appropriate to use lightness as one of the attributes of the color appearance of objects. The fundamental importance of lightness as a perceptual attribute of color is supported by the clear definition of a lightness metric in both the CIELAB and CIELUV color spaces.[2] The CIELAB and CIELUV parameter L^* is used to predict the lightness of object colors. L^* is defined in Eq. (1).

$$L^* = 116(Y/Y_n)^{1/3} - 16. \tag{1}$$

Y is the luminance or luminance factor of the object in question and Y_n is the corresponding measure of the reference white.

The L^* metric of lightness depends on the definition of $V(\lambda)$. The $V(\lambda)$ function was derived using flicker photometry and has been shown to be invalid for heterochromatic brightness matches. It has long been known that equiluminance lights of different chromaticity appear different in brightness. This effect was noted by Helmholtz and experimentally investigated by Kohlrausch.[3] Thus it has been named the Helmholtz–Kohlrausch effect. The Helmholtz–Kohlrausch effect is often stated as an increase in the brightness of equiluminance lights with increasing saturation. A great deal of research has been performed on this effect for lights viewed as unrelated, aperture colors. Pridmore[4] discusses recent work in this realm and provides many historical references. In fact, the CIE[5] has recommended a model for the prediction of heterochromatic brightness matches that is often referred to as the Ware and Cowan conversion factor. However, the CIE model is limited to predictions of the brightness of unrelated colors and therefore cannot be used to predict the lightness of object colors.

Little research has been reported on heterochromatic lightness matching of object colors. An historical review of previously published work follows to set the stage for this article and explain the origins of data used for model fitting and testing. The Helmholtz–Kohlrausch effect is redefined for object colors to state that perceived lightness increases as chroma increases for colors of equal luminance factor. Since L^* depends only on luminance factor it is not a good correlate for lightness when comparing colors of different hue and chroma. Sanders and Wyszecki[6–8] reported a series of experiments on heterochromatic lightness matching of object color. They presented observers with a test sample and then in turn asked them to choose an achromatic sample that matched it in lightness. The achromatic samples were mounted on a sliding scale behind a mask. The achromatic scale was prepared with increments of 0.5 Munsell value and observers estimated the matches to within 0.1 value steps.

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These experiments are limited in that only two or three observers were used. Therefore the data are not representative of the entire population and it is not possible to accurately estimate interobserver variability. However, the reported ability of the observers to estimate lightness matches to within 0.1 value steps implies a rough estimate of uncertainty that would correspond to 1 CIELAB L^* unit. Sanders and Wyszecki modeled their results in terms of CIE tristimulus values[6,7] and later in terms of chromaticity coordinates[8] with some degree of success. These data were used to test the models derived in this article.

Wyszecki[9] later reported results of a series of experiments that were carried out by the Committee on Uniform Color Scales of the Optical Society of America. In these experiments, 43 ceramic tiles of various chromaticity but all of approximately Munsell value 6/ were matched in lightness to a series of achromatic tiles by 76 observers. Achromatic samples were presented simultaneously with increments in Munsell value of 0.2. While the results for 76 observers provide a good estimate of the average population response, no estimates of observer variability were reported (except a comment implying that the observational uncertainty was quite high). The Wyszecki data were used to fit the models presented in this article.

There has been a recent resurgence of interest in the Helmholtz–Kohlrausch effect on object colors, perhaps instigated by the push for the development of more sophisticated and accurate color appearance models. Ikeda *et al.*[10,11] have measured the equivalent lightness of various chromatic object colors over a range of viewing illuminances and modeled the relative contributions of the scotopic and photopic systems to perceived lightness. Nayatani *et al.*[12] recently published the results of a visual experiment in which four observers made heterochromatic lightness matches to NCS samples. These results were modeled with extensions of the color appearance model proposed by Nayatani and co-workers.[13] Nayatani *et al.*[12] also modeled that Wyszecki (1967)[9] data with a linear combination of an offset and the lightness and chroma terms from their color appearance model. They reported an r^2 value of 0.82 for their fit to these data, which can be used for comparison.

There are two rather complex models of color appearance being proposed. The first is the model of Nayatani and coworkers[13] discussed above. The second model is that of Hunt.[14] These models are designed to predict the hue, lightness, brightness, colorfulness, and chroma of object colors under a wide range of illuminance levels, illuminant colors, and surround conditions. The Hunt model qualitatively predicts the Helmholtz–Kohlrausch effect since its brightness correlate depends on both achromatic and colorfulness responses and its lightness correlate depends on the brightness measure. Nayatani *et al.* have recently published an extension of their model to predict the Helmholtz–Kohlrausch effect.[12] These models require further testing with quantitative visual data. They also have some limitations in terms of practical application since the calculations required are rather complex. At this time neither model has been sufficiently tested with quantitative visual data to suggest that they predict appearance in practical applications better than simpler models.

The aim of this research has been to develop and test a CIELAB-based model capable of predicting the perceived lightness of achromatic and chromatic object colors. The aim of this model is not to represent human physiology in any way, but to relate commonly used colorimetric parameters to experimentally observed appearance attributes. The goal has been to keep the model simple enough that it can be implemented in practical circumstances in which it is necessary to compare the perceived lightness of various colors. To these ends the model is based on the CIELAB parameters L^*, C^*, and $h°$. The model was fitted using the Wyszecki (1967)[9] data and tested with the Sanders and Wyszecki (1958)[8] data and the results of a new visual experiment described in this article.

MODEL FORMULATION

The first step in the formulation of the perceived lightness model was to gather appropriate data with which to perform the statistical optimization. The Wyszecki (1967)[9] data were chosen for this purpose. The main advantages of these data were the use of object colors, the large number of observers (76) and the large number (43) and variety of test samples. The main drawbacks of these data were the lack of estimates of interobserver variability and the fact that all of the test samples were at the same Munsell value 6/, $L^* \cong 60$). The data were collected under natural or artificial daylight of at least 500 lux and reported as chromaticity coordinates and luminance factors for CIE illuminant C and the 1931 standard colorimetric observer. These data were converted to CIELAB coordinates using illuminant C tristimulus values for the normalization. Figure 1 shows the experimental data. The CIELAB L^* of the achromatic tile that matched each test sample is called the observed lightness and is plotted as a function of the measured CIELAB L^* for each tile. Since all of the tiles were nearly the same Munsell value, they all have L^*s of approximately 60. However, it can be seen in Fig. 1 that the tiles exhibited a wide range of observed lightness. Figure 1 illustrates that CIELAB L^* is not a good predictor of perceived lightness.

Examination of the difference between the observed lightness and L^* as a function of chroma provides an excellent illustration of the Helmholtz–Kohlrausch effect, as shown in Fig. 2. The large magnitude of the effect should be noted. Some samples deviate in perceived lightness from the L^* prediction by over 8 L^* units. The systematic nature of the deviations should also be noted. This systematic nature suggests that perceived lightness could be modeled by adding a linear function of C^* to the normal L^* metric. Figure 2 shows such a linear function fitted with the data. This type of model produces a metric of perceived lightness defined in Eq. (2).

$$L_1^{**} = L^* + kC^*. \tag{2}$$

L_1^{**} is the correlate to perceived lightness, L^* is the CIELAB

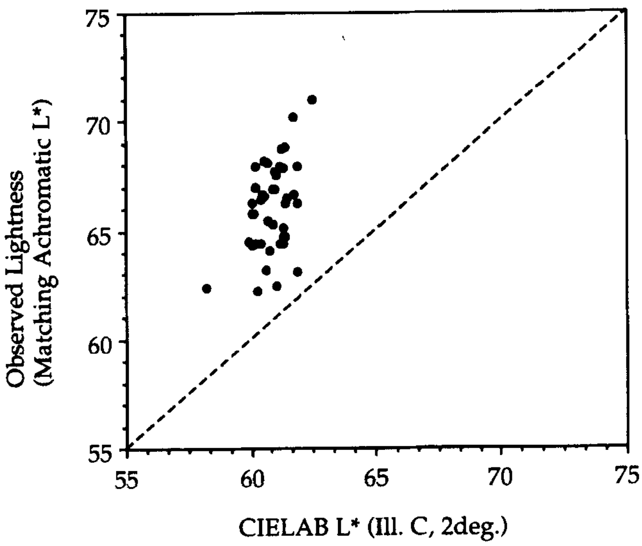

FIG. 1. The experimental results of Wyszecki (1967). The observed lightness of 43 ceramic tiles of approximately Munsell value 6/. Observed lightness is the CIELAB L^* of an achromatic tile that was a lightness match to each test tile. The RMS deviation between the observed lightness and the L^* values is 5.6.

lightness metric, C^* is the CIELAB chroma metric, and k is a constant determined by linear regression. The constant was determined to be 0.143. The regression was significant with an F value of 473, a probability > F of 0.000, and an r^2 of 0.92. A positive feature of this model is that L_1^{**} is equal to L^* for achromatic samples (i.e., $C^* = 0$). This was tested by including an offset term in the regression that was found to not be statistically significant at 95% confidence. This offset was not statistically significant in any of the models discussed in this article.

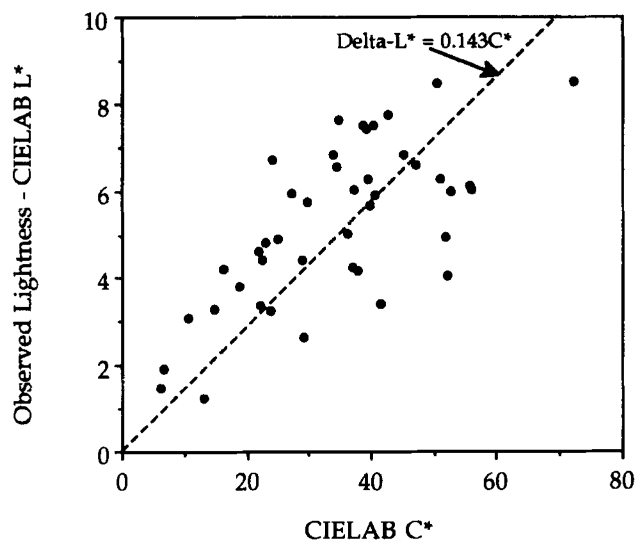

FIG. 2. The same data as shown in Fig. 1 plotted as the difference between observed lightness and CIELAB L^* as a function of CIELAB chroma C^*. These data are well fitted by a straight line of slope 0.143.

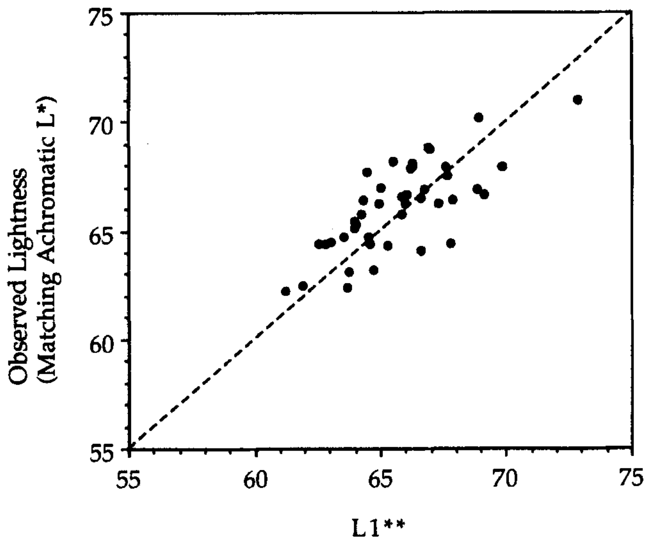

FIG. 3. The experimental results of Wyszecki (1967) plotted as a function of the L_1^{**} metric derived in the text. The RMS deviation between the observed lightness and the L_1^{**} values is 1.6.

After fitting with the Wyszecki data, the first model for perceived lightness is expressed in Eq. (3).

$$L_1^{**} = L^* + 0.143C^*. \tag{3}$$

Figure 3 shows the Wyszecki data plotted as a function of L_1^{**}. It is clear that the systematic error in the L^* predictions has been accounted for to a high degree. The RMS deviation between the L_1^{**} predictions and the experimental results is 1.6 CIELAB units. This is slightly larger than the intraobserver uncertainty estimated from the studies of Sanders and Wyszecki.

It has often been reported (e.g., Ref. 11), especially in the literature on heterochromatic brightness matching of unrelated colors, that the Helmholtz–Kohlrausch effect has a hue dependency. It is normally stated that the effect is not as large for yellow hues as it is for others. The differences between observed lightness and L^* were examined as a function of hue angle at various levels of chroma (in order to eliminate the obvious chroma dependency). These examinations indicated that there was a strong trend whereby the effect dropped off at a CIELAB hue angle of 90° (i.e., yellow). While there were not enough data at constant-chroma levels to fit an analytical function to the hue dependency, the curve shape was selected to be $\frac{1}{2}$-cycle of the absolute value of a sinusoid with a minimum at $h° = 90°$. This function is plotted in Fig. 4 in both rectangular and polar coordinates. The form of this hue dependency has been corroborated in Ikeda *et al.*'s recent article.[11] The L_1^{**} model was enhanced to include the dependency on hue angle by replacing the constant factor k with a function of $h°$ of the type illustrated in Fig. 4. Equation (2) becomes Eq. (4) by replacing k with $f1(h°)$:

$$L_2^{**} = L^* + f1(h°)C^* \tag{4}$$

where

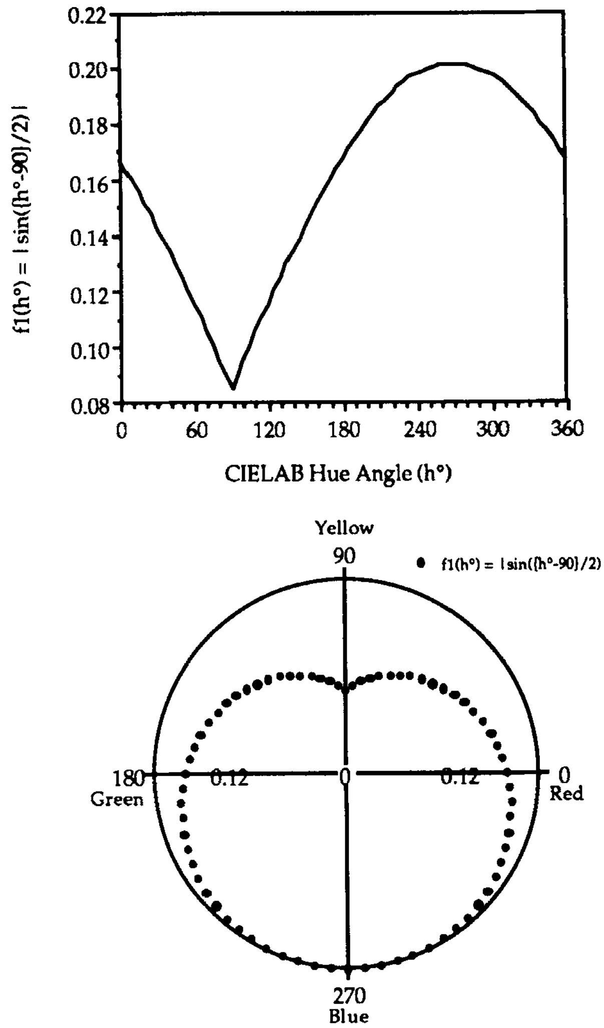

FIG. 4. The $f1(h°)$ function that describes the magnitude of the Helmholtz–Kohlrausch effect as a function of CIELAB $h°$ plotted on both rectangular and polar coordinates.

$$f1(h°) = k_1\left|\sin\left(\frac{h° - 90}{2}\right)\right| + k_2. \tag{5}$$

The two constants in the L_2^{**} model were determined to be 0.116 and 0.085 via linear regression. Both terms were statistically significant. The model had an F value of 611, probability > F of 0.000, and an r^2 of 0.97. Note that with these coefficients the k term that was a constant in the L_1^{**} model varies from 0.085 to 0.201, depending on $h°$, in the L_2^{**} model. With the regression coefficients included, the L_2^{**} model is expressed in Eqs. (6) and (7).

$$L_2^{**} = L^* + f1(h°)C^*, \tag{6}$$

$$f1(h°) = 0.116\left|\sin\left(\frac{h° - 90}{2}\right)\right| + 0.085. \tag{7}$$

Figure 5 shows the Wyszecki data plotted as a function of L_2^{**}. The systematic error in the L^* predictions has been accounted for to a higher degree than with the L_1^{**} model. The RMS deviation between the L_2^{**} predictions and the experimental results is 1.0 CIELAB unit. This equals the intraobserver uncertainty estimated from the studies of Sanders and Wyszecki. Thus it would be of no value to make the model any more complex since it would then be modeling the noise in the data.

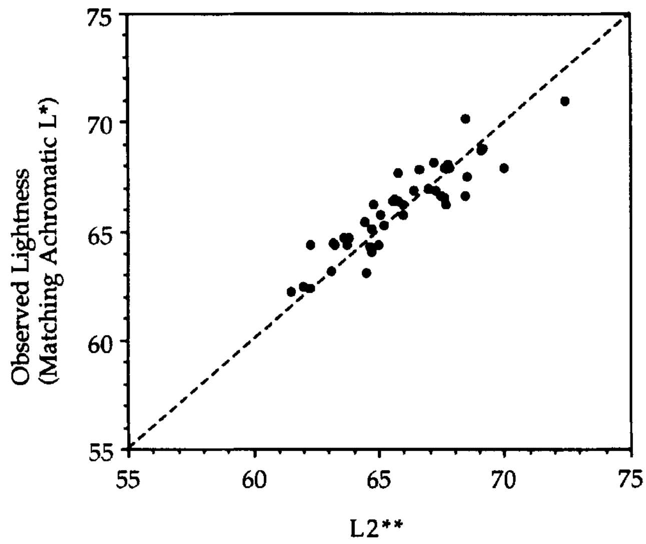

FIG. 5. The experimental results of Wyszecki (1967) plotted as a function of the L_2^{**} metric derived in the text. The RMS deviation between the observed lightness and the L_2^{**} values is 1.0.

Given well fitting, simple models that meet the intended objectives, the next step was to test them with independent data. The Sanders and Wyszecki (1958)[8] data were used for this purpose. In these experiments, three observers matched the lightness of 106 samples of various chromaticity and luminance factor to a series of achromatic samples. The experiment was performed under MacBeth daylight at approximately 650 lux. One advantage of these data are the large number of samples over a range of luminance factors. Sanders and Wyszecki concluded from these data that the ratio of lightness to luminance factor was constant for various luminance factors. Since the L^* cube root approximates a logarithmic transform this would suggest that the difference between observed lightness and L^* would be constant for various L^* values. These data allow a test of this assumption. However, only three observers were used and the interobserver variance was at times larger than the effect being measured. The mean interobserver standard deviation of the Sanders and Wyszecki data was 1.7 CIELAB units, while the mean effect was 0.9 CIELAB units.

L_1^{**} predicted the Sanders and Wyszecki data with an RMS deviation of 2.7 CIELAB units. The RMS deviation for the L_2^{**} predictions was 1.4 CIELAB units. These results indicate that the models are robust and capable of predicting other data to well within the experimental uncertainty. However, the small number of observers and large variability in the data leave the performance of the models incompletely tested. In addition, the number of samples lighter or darker than medium lightness was somewhat limited, leaving open the question of independence of the effect with respect to luminance factor.

With these limitations in mind, it was decided that further data were required to test the models. Due to the paucity of published results using object colors, a new experiment was designed and executed. This experiment is described in the following sections.

VISUAL EXPERIMENT

A visual experiment was performed in which 11 observers matched the perceived lightness of a continuously variable achromatic sample to the perceived lightness of 36 Munsell papers. Munsell samples were chosen at three values (3/, 5/, and 7/), four hues (5R, 5Y, 2.5G, and 5PB), and three chroma levels (low, medium, and high). The samples were glossy painted-paper cut to 7.2-by-9.6 cm^2. Their actual Munsell designations are listed in Table I. Each sample was mounted on a 10.0-by-15.5 cm^2 gray card of Munsell value 5/.

Samples were viewed one at a time in an ACS Datacolor VCS-10 under fluorescent simulated daylight (6900 K, 1200 lux). The VCS-10 is a Maxwell-disk device that allows an observer to interactively create additive mixtures of glossy materials. Even when spinning at rates necessary for visual fusion, the VCS-10 stimulus retains its glossy, surface-color appearance. The VCS-10 stimulus was masked to 7.2-by-9.6 cm^2. The back surface (stimuli surround) of the VCS-10 was covered with N5/ cardboard. The test samples were placed adjacent to the VCS-10 stimulus with an N5/ separation of 4 mm. The samples were viewed from a distance of 50 cm. Thus the stimuli together subtended a visual angle of 16° by 10°. A 6-by-8 cm^2 reference white patch was placed 7 cm above the samples to meet the conditions required by the definition of lightness. The viewing geometry was approximately 45° illumination with normal (0°) viewing. The observers placed each sample in the viewing booth and then adjusted the white and black controls of the VCS-10 until a lightness match was made. These VCS-10 controls allowed the observers to produce a gray of any luminance factor between 0.0037 and 0.9110. The observers recorded the white and black VCS-10 readings for the lightness match to each sample. Each observer repeated the experiment three times on different days. The following instructions were given:

> There are 36 samples. For each sample you are asked to make a lightness match using the VCS. Lightness is defined as **the brightness of an area judged relative to the brightness of a similarly illuminated area that appears white.** To begin, place the first sample on the background under the holding band. Make sure the bottom edge of the card is flush against the stop. Please be sure not to touch the sample. Manipulate only the white +/− switch to make a lightness match. Utilize the fast/slow switch as necessary, finalizing each match in slow speed. When a match is reached, record the values of black (B) and white (W) in the appropriate columns on the answer sheet. Remove the finished sample and replace it with the next. Please go in order (check number on back of card).

TABLE I. Results of the visual experiment. "Observed lightness" is the mean L^* value of an achromatic sample that appeared the same lightness as the test sample. "Std. dev." is the interobserver standard deviation of the observed lightness ($n = 11$). The mean standard deviation was 4.2. The average intraobserver standard deviation for three replicate trials was 3.5.

No.	Munsell notation	L^*	C^*	$h°$	Observed lightness	Std. dev.
1	5R3/2	30.58	13.68	25.46	38.1	2.9
2	5R3/6	31.19	33.33	25.34	39.1	4.1
3	5R3/10	30.89	53.40	24.60	38.8	4.7
4	5R5/2	51.64	9.95	26.19	55.2	2.3
5	5R5/8	51.41	40.99	27.15	56.7	4.6
6	5R5/14	52.00	71.23	29.40	62.9	10.9
7	5R7/2	72.49	8.72	28.37	71.4	4.6
8	5R7/6	71.56	29.51	28.29	72.1	2.8
9	5R7/10	72.60	48.64	27.81	75.0	6.2
10	5Y3/1	30.34	8.30	90.99	36.3	2.8
11	5Y3/2	30.30	13.97	89.36	36.4	3.1
12	5Y3/4	30.58	28.50	87.52	35.4	4.2
13	5Y5/2	51.52	15.87	93.54	57.0	1.6
14	5Y5/6	51.43	47.03	89.96	48.6	4.2
15	5Y5/8	51.59	59.71	88.42	50.9	5.4
16	5Y7/2	71.77	16.67	95.07	71.4	2.8
17	5Y7/8	71.82	59.83	91.06	65.5	4.9
18	5Y7/12	71.69	86.70	89.59	66.1	8.4
19	2.5G3/2	30.22	10.95	151.02	36.4	3.0
20	2.5G3/6	30.81	34.31	155.69	39.2	4.3
21	2.5G3/10	30.91	55.44	156.91	40.9	5.1
22	2.5G5/2	51.20	12.60	149.41	56.0	2.0
23	2.5G5/8	52.11	48.61	152.67	59.5	3.3
24	2.5G5/12	50.81	65.89	154.59	61.5	5.5
25	2.5G7/2	72.79	13.55	148.52	71.3	3.5
26	2.5G7/6	72.16	37.75	150.32	73.4	2.3
27	2.5G7/10	71.32	58.28	152.88	70.0	4.8
28	5PB3/2	30.27	9.99	275.89	38.4	3.3
29	5PB3/6	30.41	27.82	274.26	44.4	6.4
30	5PB3/10	30.42	44.05	276.56	48.6	9.3
31	5PB5/2	51.82	7.20	273.99	57.1	1.8
32	5PB5/8	52.05	31.87	272.63	61.4	3.6
33	5PB5/12	51.14	46.26	270.92	62.8	6.0
34	5PB7/2	72.08	6.05	266.63	75.4	1.6
35	5PB7/6	71.87	22.02	267.04	73.3	2.3
36	5PB7/8	71.17	28.68	266.47	74.7	1.9

The samples were presented in a fixed random order. Each observer required between 15 and 45 minutes to complete a session (36 matches).

The VCS-10 readings were converted into luminance factor and then to CIELAB L^* via an instrumentally determined calibration function. The VCS-10 stimuli were measured *in situ* using its light source and a PhotoResearch PR-703A spectroradiometer in the position of the observers. A pressed polytetrafluoroethylene (PTFE) sample was used as a reference white with a 45/0 luminance factor of 1.00. The VCS-10 black had a measured reflectance factor of 0.0037 and the white a reflectance factor of 0.9110. Since the VCS-10 relies on additive mixture of the white and black to produce the intermediate grays, the luminance factors of

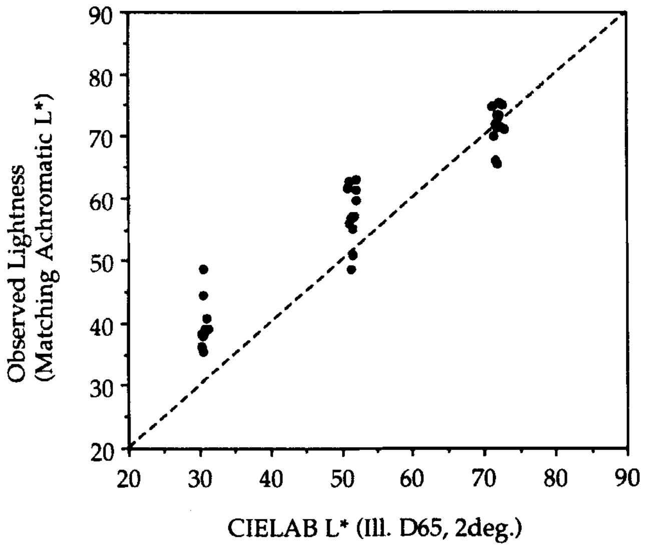

FIG. 6. Results of the experiment described in this article in which 11 observers made lightness matches to 36 Munsell samples. Observed lightness is the CIELAB L^* of an achromatic stimulus that was a lightness match to each test sample. The RMS deviation between the observed lightness and the L^* values is 7.1. The interobserver standard deviation was 4.2.

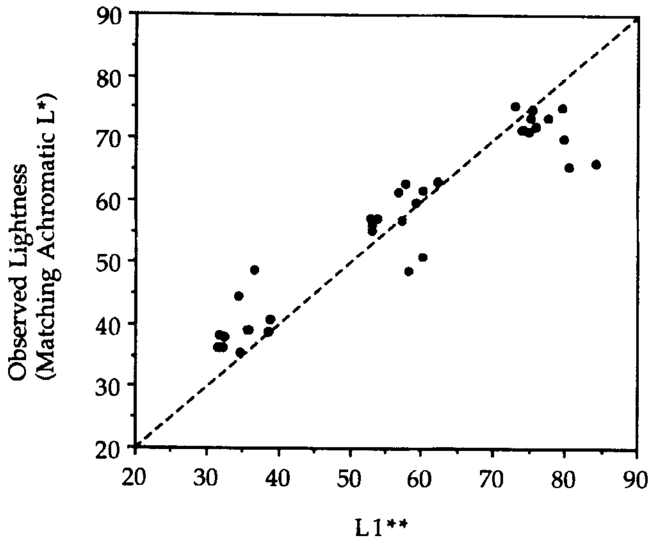

FIG. 7. The new experimental results plotted as a function of the L_1^{**} metric derived in the text. The RMS deviation between the observed lightness and the L_1^{**} values is 6.2.

the grays are linear with VCS-10 reading. This was verified by measuring the luminance factors of three intermediate gray mixtures. The linearity was exact. Since the minimum white reading was 0 and the maximum was 191, the conversion from VCS-10 reading (VCS) to CIELAB L^* is given by Eq. (8).

$$L^* = 116[0.0037 + 0.00475(\text{VCS})]^{1/3} - 16. \quad (8)$$

The Munsell test samples were measured on a Milton Roy ColorMate 45/0 spectrophotometer. Their CIELAB coordinates were calculated using the 1931 CIE standard colorimetric observer (2°) and CIE standard illuminant D65. These coordinates are also reported in Table I.

The 11 observers were drawn from a population of college students and staff. They had a mean age of 28 with a range from 22 to 34. All observers had normal color vision. This was verified with a set of pseudoisochromatic plates. Most of the observers had experience in color matching and all were experienced in psychophysical experiments.

RESULTS AND DISCUSSION

The experimental results are expressed in terms of the CIELAB L^* of the achromatic stimulus that matched the test stimulus in perceived lightness. These values are referred to as the observed or perceived lightness of the samples. It should be noted that these measurements are still in CIELAB units. The average results and the interobserver ($n = 11$) standard deviations are listed in Table I. The mean results are plotted in Fig. 6. The RMS deviation between the observed lightness and the CIELAB L^* prediction is 7.1. The systematic nature of the deviations is also evident in Fig. 6. The intraobserver ($n = 3$) standard deviations ranged from 1.9 to 5.4 with a mean of 3.5. The mean interobserver ($n = 11$) standard deviation was 4.2. This standard deviation was taken as the goal for the RMS deviation between the model predictions and the experimentally observed lightness. This would mean that the model predicts lightness as a member of the population of normal observers and within 1 standard deviation of the mean values.

L_1^{**} is a substantially better predictor of the experimental results than L^*. Figure 7 shows the L_1^{**} predictions of the experimental data. The RMS deviation between the observed lightness and the L_1^{**} predictions is 6.2. This is a small improvement over the L^* predictions. However, as can be seen in Fig. 7, most of the systematic nature of the deviations has been modeled. It is worth noting that there appears to be a dependency on overall lightness. The effect is larger than predicted for the value 3/ samples and smaller than predicted for the value 7/ samples. The prediction for the value 5/ samples is excellent. This is not surprising since the models were fitted to experimental data gathered with samples of Munsell value 6/ and no reliable data at other luminance factors were available.

Similar results for the L_2^{**} model are shown in Fig. 8. The RMS deviation between the observed lightness and the L_2^{**} predictions is 5.2. This is an improvement over the L_1^{**} model as expected. The RMS deviation of 5.2 is still larger than the interobserver standard deviation in the experimental results. However, it should be noted that the total uncertainty in the experimental results obtained by quadrature summation of the interobserver and intraobserver standard deviations is 5.5 CIELAB units. Therefore the L_2^{**} model makes predictions within the total uncertainty of the data, but not within the interobserver uncertainty. While most of the systematic departures from L^* are predicted by the L_2^{**} model, it is clear in Fig. 8 that the effect

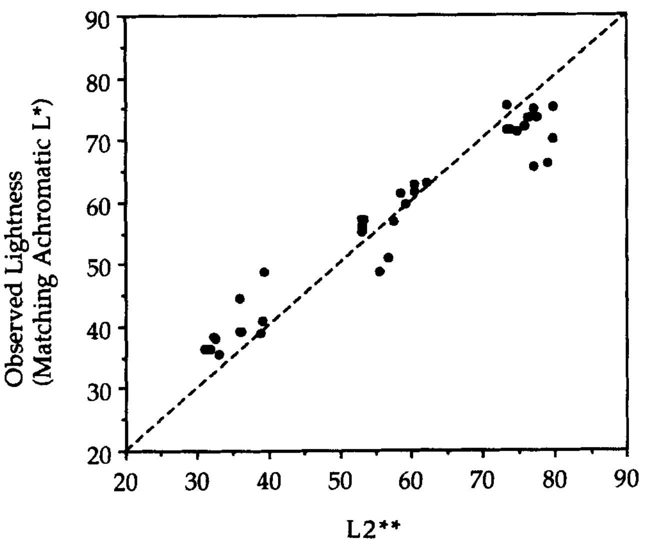

FIG. 8. The new experimental results plotted as a function of the L_2^{**} metric derived in the text. The RMS deviation between the observed lightness and the L_2^{**} values is 5.2.

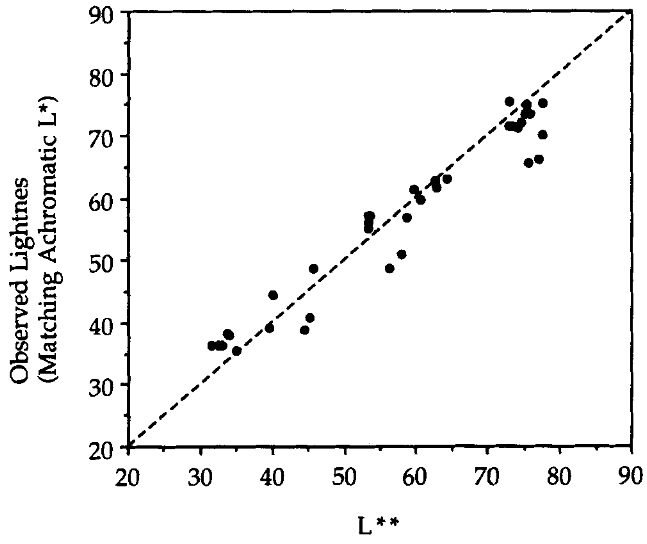

FIG. 9. The new experimental results plotted as a function of the L^{**} metric derived in the text. The RMS deviation between the observed lightness and the L^{**} values is 4.2.

is greater for value 3/ samples and lesser for value 7/ samples.

While the L_1^{**} and L_2^{**} models are substantially better than L^*, they can be improved. None of the previously published results included significant data over a range of luminance factors. The L_1^{**} and L_2^{**} models were derived under the commonly held assumption that the lightness-to-luminance-factor ratio for a given chromaticity is constant across changes in luminance factor. The data reported in this article fail to corroborate that assumption. It is clear that the Helmholtz–Kohlrausch effect (measured in L^* units) is much larger at low luminance factors than at high luminance factors. Therefore it is justified to add a simple improvement to the model to predict this result, which was also confirmed by comments from the observers following the experiments.

The addition of a multiplicative factor that is a function of lightness results in a metric of perceived lightness defined in Eq. (9).

$$L^{**} = L^* + f2(L^*)f1(h^\circ)C^*. \tag{9}$$

The $f2(L^*)$ function is defined at two points. It was set equal to 1.0 at $L^* = 60$ so that the model fit to the original Wyszecki (1967)[9] data would be unchanged and it was set equal to 0.0 at $L^* = 100$, indicating that the effect is nonexistent for a perfect white. Given only two predefined points, there is no justification in using a function of any higher order than linear. Equation (10) is the linear function that passes through these points.

$$f2(L^*) = 2.5 - 0.025L^*. \tag{10}$$

The hue-angle function, $f2(h^\circ)$, remains unchanged and is given by Eq. (7). Thus Eq. (9) [with the functions defined in Eqs. (7) and (10)] is the final formulation of the new metric for perceived lightness, L^{**}.

Figure 9 shows the L^{**} predictions of the experimental data. The RMS deviation between the observed lightness and the L^{**} predictions is 4.2. This is a significant improvement over the L^* predictions. The predictions are also substantially better than those of the first two forms of the model. The systematic nature of the deviations most notable in the L^* predictions is not apparent in the L^{**} predictions. L^{**} is a substantially better predictor of the experimental results than L^*. The RMS deviation for the L^{**} metric is the same as the interobserver standard deviation. Therefore this model meets the criterion of being within the interobserver uncertainty.

Figure 10 shows the deviation between the mean perceived lightness and the L^{**} prediction as a function of the experimentally determined interobserver standard deviation for each sample. Points falling between the dotted lines are within 1 standard deviation, points between the dashed lines are within 2 standard deviations, and points between the solid lines are within 3 standard deviations. Figure 10 shows that the majority of the predictions fall within 2 standard deviations of the mean value and that there is little or no systematic error in the predictions.

Since the L^{**} metric accounts for all of the apparent systematic departures of perceived lightness from L^* and its RMS deviation is the same as the interobserver standard deviation, there is no reason to formulate a more complex model. Further experiments should be performed to test the robustness of this model to changes in illuminance level, illuminant color, surround luminance, surround color, sample size, sample separation, and any other experimental parameters that might influence the results.

CONCLUSION

A new metric for lightness, L^{**}, defined by Eqs. (9), (7), and (10), has been formulated. This metric predicts the

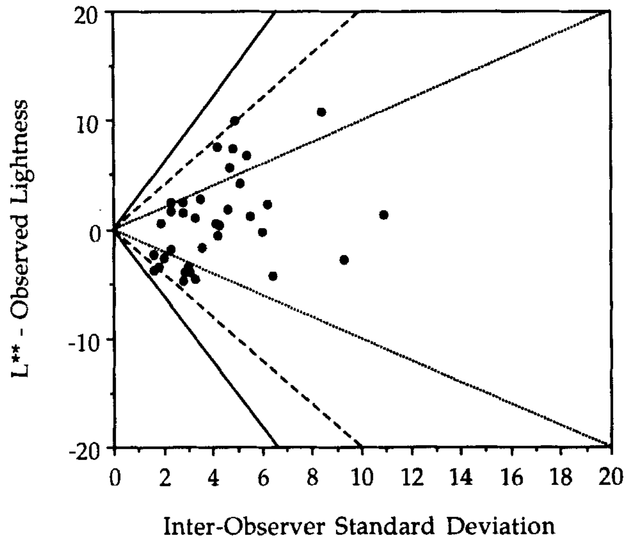

FIG. 10. The deviation between observed lightness and the L^{**} predictions as a function of the experimental interobserver standard deviation ($n = 11$) for each sample. Points falling within the dotted lines indicate predictions that are within 1 standard deviation of the mean, points falling within the dashed lines are within 2 standard deviations, and points falling within the solid lines are within 3 standards deviations.

perceived lightness of all object colors within the experimentally determined interobserver uncertainty. Those wishing to use the CIELAB color space as a first-order description of object-color appearance should use the L^{**} metric as the correlate of lightness rather than the L^* metric. Those wishing to use an even simpler model might consider the L_1^{**} and L_2^{**} metrics derived in this article. Their predictions are not as accurate as the L^{**} metric, but they represent a significant advance over the L^* metric.

The L^{**} metric is very easy to implement. The first step to calculate the CIELAB L^*, C^*, and $h°$ parameters in the usual way. Then the $f1(h°)$ and $f2(L^*)$ values are calculated using Eqs. (7) and (10). Finally, the L^{**} metric is calculated using Eq. (9).

L^{**} is an extension of a recognized and highly utilized international standard. Therefore it represents a minimum investment in programming and education for its implementation. It would be very easy for those who are currently using CIELAB to begin using L^{**} as a lightness metric.

It may also be advantageous to use the L^{**} metric rather than L^* in color-difference calculations. This would be useful when comparing a standard and sample that are at different chroma levels. If they had identical L^* values the lightness contribution to the color difference would be 0. However, since they are at different chroma, the two would appear to be of different lightness. The L^{**} metric would reflect this difference and therefore may be more suitable for the color-difference calculation. The use of L^{**} does not provide significant improvements when small color differences are being calculated. However, when very large color differences are being calculated, such as those found in the color reproduction industry, the L^{**} metric might provide significant improvement in the correlation between visual and instrumental color differences.

The performance of the L^{**} metric compares favorably with results obtained using more complex color appearance models. For example, Nayatani *et al.*[12] fitted a model that combined lightness and chroma terms from their appearance model to the Wyszecki (1967)[9] data. Their model was based on a color appearance model that is substantially more complex than CIELAB and they added three more free parameters to fit the Wyszecki data. They reported an r^2 goodness-of-fit metric of 0.82. The L^{**} metric described in this article has only two free parameters and results in an r^2 value of 0.97. This illustrates that a simple CIELAB-based model can perform as well as, or better than, a more complicated model. Perhaps enhancements of currently recommended color spaces that can be used for the prediction of color appearance should be given serious consideration. There is not enough visual data available to justify the adoption of more complex color appearance models at this time. In addition, the very large interobserver variability found in color appearance judgements does not warrant complex appearance models when simpler models are capable of predicting the results within the uncertainty. Another example of this approach to modeling appearance has been given by Hunt[15] in which correlates of lightness based on CIELAB L^* are proposed for conditions in which the surround is dim or dark. This type of approach deserves further investigation.

Finally, this article includes an additional color appearance data set in Table I that can be used by others to develop and test color appearance models.

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